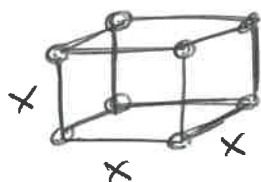


# Formula Booklet - Sat Math

1

## (1) Surface Area (Δεν δίνονται)

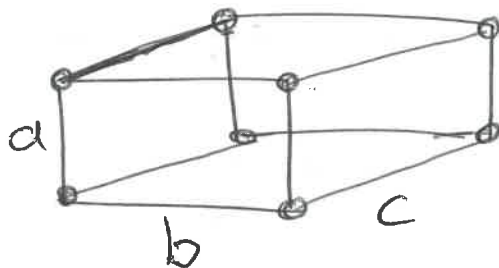
Cube



(όλες οι όψεις ίσες μεταξύ τους)

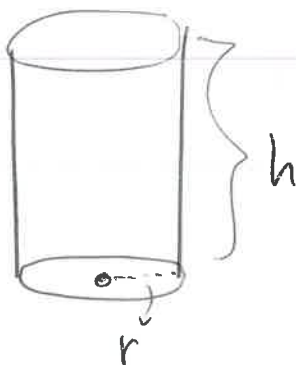
$$S = 6 \cdot s^2$$

Prism



$$S = 2 \cdot (ab + bc + ac)$$

Cylinder



$$S = 2\pi r \cdot h + 2\pi r^2$$

Sphere



$$S = 4\pi r^2$$

(2)

Άσκηση

Two identical rectangular prisms each have a height of 50. The base of each prism is a square, the surface area of each prism is  $K \text{ cm}^2$ . If the prisms are glued together along a square base, the resulting prism has a surface area of  $\frac{11}{6} K \text{ cm}^2$ . What is the side length in cm, of each square base?

Λύση:

•  $h$    $h = 50$   $\Rightarrow$  Surface Area  $= 2s^2 + 4 \cdot h \cdot s = 2s^2 + 200s = K$  ~~(1)~~

• Κολλάω 2 πρίσματα along the base και παίρνω  $\frac{11}{6} K$  και

συνάγω όσα από  $2 \cdot K = \frac{12}{6} K$  και  $\frac{12}{6} \cdot K - \frac{11}{6} \cdot K = \frac{1}{6} \cdot K$ ,

"χάνω"  $\frac{1}{6} K$ , άρα 2 βάσεις (ως προς το εμβαδόν (area)) ουσιαστικά το χάνουν κολλημένα.

• Άρα  $2 \cdot s^2 = \frac{1}{6} \cdot K$  ~~(1)~~  $\Rightarrow 2s^2 = \frac{1}{6} (2s^2 + 200s) \Rightarrow$

$$\Rightarrow 12s^2 - 2s^2 - 200s = 0 \Rightarrow$$

$$\Rightarrow 10s(s - 20) = 0 \stackrel{s \neq 0}{\Rightarrow}$$

$$\Rightarrow \boxed{s = 20 \text{ cm}}$$

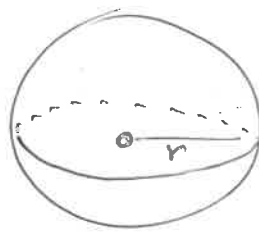
(2) Volumes

Cube  $V = x^3$  (για κύβο πλευράς  $x$ )

Prism  $V = a \cdot b \cdot c$

Cylinder  $V = \pi \cdot r^2 \cdot h$

Sphere



$$V = \frac{4}{3} \pi \cdot r^3$$

(3) Sum of Angles

Triangle |  $180^\circ \underline{=} \pi \text{ rad}$

Square/Parallelogram |  $360^\circ \underline{=} 2\pi \text{ rad}$

General shape with  $n$ -angles |  $(n-2) \cdot 180^\circ \underline{=} (n-2) \cdot \pi \text{ rad}$

⊛ Regular Polygon :

All sides/angles are equal to each other.

Σημειώσεις

Ομοιότητα Σχημάτων (Similar Shapes)

2D-Shapes Let  $A, B$ : similar shapes (of the same type)  
 (two triangles, two squares etc)

If  $\lambda$  = similarity ratio, then:

$$\frac{\text{Per}(A)}{\text{Per}(B)} = \lambda \quad (\text{Per} = \text{περίμετρος})$$

$$\frac{\text{Area}(A)}{\text{Area}(B)} = \lambda^2 \quad (\text{Area} = \text{εμβαδόν})$$

3D-Shapes

Let  $A, B$ : similar shapes (of the same type).

If  $\lambda$  = similarity ratio, then:

$$\frac{\text{Area}(A)}{\text{Area}(B)} = \lambda^2$$

$$\frac{\text{Vol}(A)}{\text{Vol}(B)} = \lambda^3 \quad (\text{Vol} = \text{volume} = \text{όγκος})$$



Άσκηση

| <u>Sphere</u> | <u>Volume</u>     |
|---------------|-------------------|
| Sphere A      | $26244 \cdot \pi$ |
| Sphere B      | $36 \cdot \pi$    |

$$\left\{ \begin{array}{l} S_A = a \cdot \pi \\ S_B = b \cdot \pi \\ \text{Find } |a-b| \end{array} \right. \quad \left\{ \begin{array}{l} \text{surface areas} \end{array} \right.$$

$$S = 4\pi r^2$$

Solution:

\* Βρίσκω τον λόγο των ακτίνων  $r_A, r_B$ :

$$\frac{V_A}{V_B} = \frac{26244 \cdot \pi}{36 \cdot \pi} = 729 = 9^3 \Rightarrow$$

$$\Rightarrow \boxed{\lambda = 9} \quad \text{similarity ratio (Η σφαίρα A έχει 9-πλάσια ακτίνα απ' τη σφαίρα B)}$$

$$\Rightarrow \frac{S_A}{S_B} = \lambda^2 = 81 \Rightarrow S_A = 81 \cdot S_B$$

Η σφαίρα A έχει 81-πλάσιο εμβαδόν απ' τη σφαίρα B

$$\text{Άρα: } \left\{ \begin{array}{l} S_B = 4\pi \cdot r_B^2 = (4 \overset{b}{\uparrow} r_B^2) \cdot \pi \\ S_A = 4\pi \cdot r_A^2 = (4 \underset{\hookrightarrow a}{r_A^2}) \cdot \pi \end{array} \right. \Rightarrow$$

$$|a-b| = |4r_A^2 - 4r_B^2| =$$

$$= 4 \cdot |r_A^2 - r_B^2| = 4 \cdot 720 = \underline{2880}$$

$$\bullet V_A = \frac{4}{3} \pi r_A^3 \Rightarrow$$

$$26244\pi = \frac{4}{3} \pi r_A^3 \Rightarrow \underline{r_A = 27}$$

$$\bullet V_B = \frac{4}{3} \pi r_B^3 \Rightarrow$$

$$36\pi = \frac{4}{3} \pi r_B^3 \Rightarrow \underline{r_B = 3}$$

#### (4) Parabola (Quadratic Equation) :

$$y = ax^2 + bx + c \text{ (Standard form) } (2\star)$$

$$(If D > 0) y = a(x - x_1) \cdot (x - x_2) \text{ (Factorised form) } (2\star)$$

$$(If D = 0) y = a(x - x_0)^2 \text{ (Vertex form) } (3\star)$$

$$\bullet D = b^2 - 4ac \text{ (Discriminant)}$$

$$\bullet If D > 0 \Rightarrow \begin{cases} x_1 = \frac{-b + \sqrt{D}}{2a} \\ x_2 = \frac{-b - \sqrt{D}}{2a} \end{cases}$$

$$\bullet If D = 0 \Rightarrow x_0 = -\frac{b}{2a}$$

Παρατήρηση  $y = f(x) = ax^2 + bx + c$  και οι δύος της  
 εξισώσεις (roots of the equation)  $f(x) = 0$  είναι τα  
 x-intercepts της παραβολής.

If  $a > 0$   $\Rightarrow f(x)$ : concaves upwards 

If  $a < 0$   $\Rightarrow f(x)$ : concaves downwards 

Vertex:  $V(h, k)$

$h$ :  $x$ -coordinate of the vertex

$k$ :  $y$ -coordinate of the vertex

Vertex = κορυφή παραβολής

Av  $a > 0 \Rightarrow \text{minimum}$

Av  $a < 0 \Rightarrow \text{maximum}$

Vertex form:

$$y = a(x-h)^2 + k \quad (\text{υπάρχει ανεξαρτήτως του προσήτου της } D)$$

(4\*)

• 0, (1\*), (2\*), (3\*), (4\*) είναι οι τιμές κορφής της παραβολής.

• Σημειώσεις:

(\*) Η ευθεία  $x = -\frac{b}{2a}$  είναι ο άξονας συμμετρίας (axis of symmetry) της παραβολής.

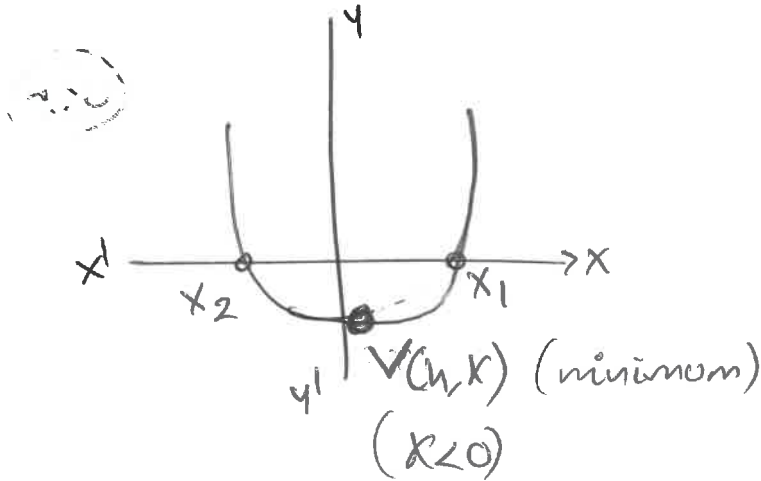
(\*) Sum of roots  $= x_1 + x_2 = -\frac{b}{a}$   
 Product of roots  $= x_1 \cdot x_2 = \frac{c}{a}$  } Vieta formulas (If  $D \geq 0$ )

(9)

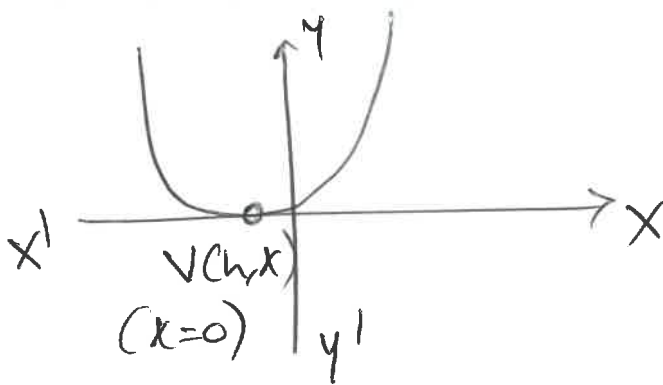
(\*) Στην μορφή  $y = ax^2 + bx + c$ , το σημείο  $(0, c)$  είναι το  $y$ -int της παραβολής (βάλε όταν  $x=0$ )

Γραφικά

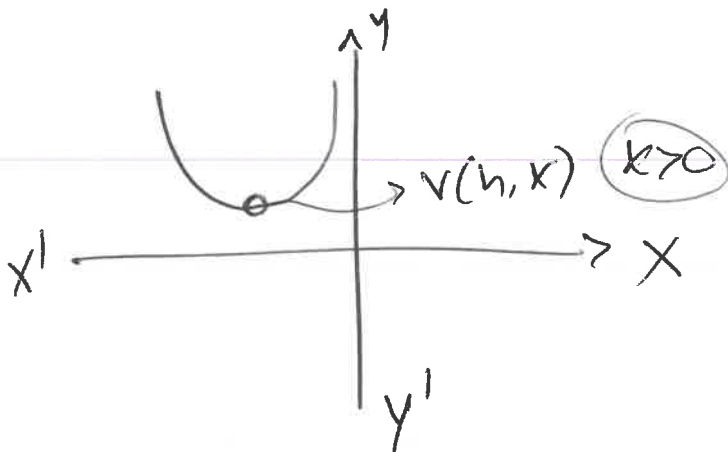
Για  $a > 0$ , έχουμε τις ακόλουθες περιπτώσεις:



$$\boxed{D > 0}$$



$$\boxed{D = 0}$$



$$\boxed{D < 0}$$

Άσκηση:

- parabola
- vertex at the point  $V(-2, 5)$
- intersects  $x$ -axis at two points

• If  $y = ax^2 + bx + c$ , what is a possible value of  $a+b+c$  is written as:

Possible Answers

$a+b+c$ ?

- (A) 2
- (B) 6
- (C) 8
- (D) 11

Λύση

$$V(-2, 5) \xrightarrow[\text{form}]{\text{Vertex}} y = a(x-h)^2 + k \xrightarrow[h=5]{h=-2}$$

$$\Rightarrow y = a(x+2)^2 + 5 = ax^2 + (4a)x + (4a+5)$$

$$\Rightarrow y = (a)x^2 + (4a)x + (4a+5) \quad (\text{and Vertex form})$$

$$\text{or } y = (a)x^2 + (b)x + c \quad (\text{and Standard form})$$

Άρα  $a+b+c = a+4a+4a+5 = 9a+5$  (1\*)

Επίσης  $k < 0$  και  $D > 0$  (intersect at 2 points)  $\Rightarrow$

$$\Rightarrow a \neq 0. \text{ Άρα } (1*): 9a+5 \neq 5$$

Η λύση και η απάντηση είναι το (A).

(5) Formulas / Identities (Algebraic Shortcuts)

$$(a+b)^2 = a^2 + 2ab + b^2$$

Square of Sum

$$(a-b)^2 = a^2 - 2ab + b^2$$

Square of Difference

$$a^2 - b^2 = (a-b)(a+b)$$

Difference of Squares

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Cube of Sum

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Cube of Difference

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

Sum of Cubes

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Difference of Cubes

# (6) Lines / Linear Equations

$$y = f(x) = m \cdot x + b \quad \text{Linear Equation}$$

$$y = m \cdot x + b \quad (\text{slope-intercept form})$$

$$a \cdot x + b \cdot y = c \quad \left\{ \begin{array}{l} \text{Standard forms} \\ a \cdot x + b \cdot y + c = 0 \end{array} \right.$$

$$\rightarrow y = \underbrace{\left(-\frac{a}{b}\right)}_{\text{slope}} x + \underbrace{\left(\frac{c}{b}\right)}_{\text{y-intercept}}$$

If  $m > 0$  :  $f$  linearly increases

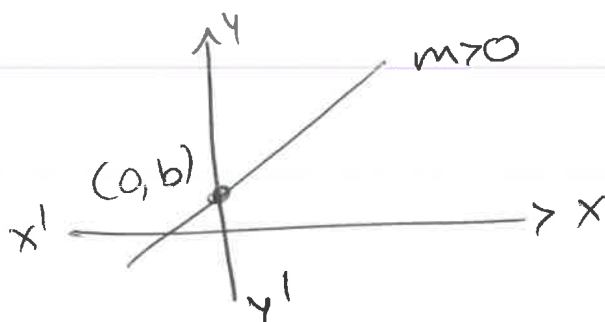
If  $m < 0$  :  $f$  linearly decreases

$$m = \text{slope} \left( \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \right)$$

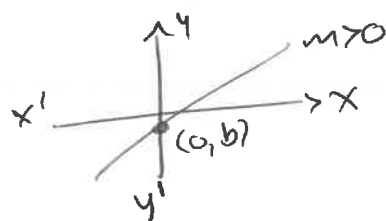
$$b = \text{y-intercept} (x=0)$$

Graphical

$$\underline{m > 0, b > 0}$$

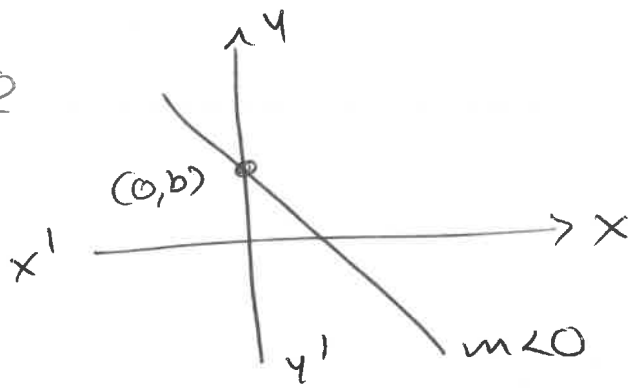


$$\underline{m > 0, b < 0}$$



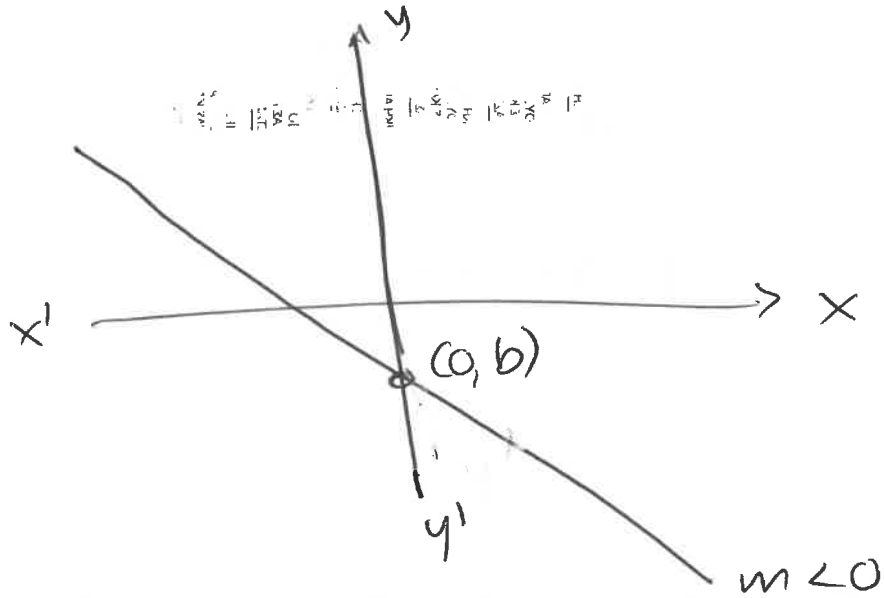


$m < 0, b > 0$



13

$m < 0, b < 0$



## (7) Systems of Linear Equations

(14)

$$\begin{cases} a_1 x + b_1 y = c_1 \\ a_2 x + b_2 y = c_2 \end{cases} \quad (\text{Standard form})$$

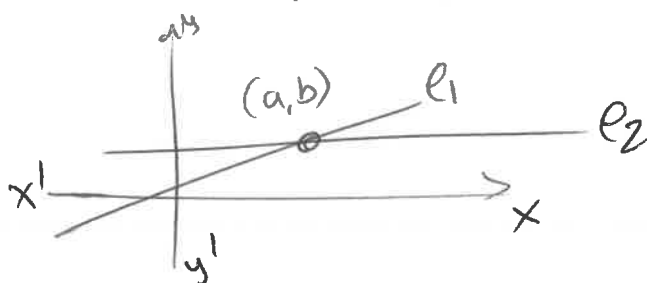
$$\begin{cases} y = m_1 \cdot x + b_1 \\ y = m_2 \cdot x + b_2 \end{cases} \quad (\text{Slope-Intercept form})$$

Περίπτωση 1 | Μοναδική Λύση / Unique Solution

Οι ευθείες τέφνονται σε σημείο  $(a, b)$ .  
(intersect)

Το  $(a, b)$  είναι η Λύση του συστήματος.  
(solution)

Πρέπει  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$  (i)  $m_1 \neq m_2$



( $l_1$  : line 1)  
( $l_2$  : line 2)

## Περίπτωση 2

Καμία λύση

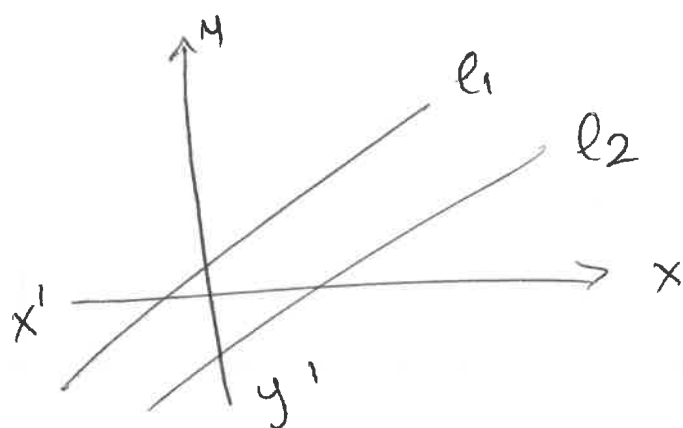
Αδύνατο Σύστημα  
Παράλληλες Εξισώσεις  
(Parallel Lines)

15

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \quad (\neq)$$

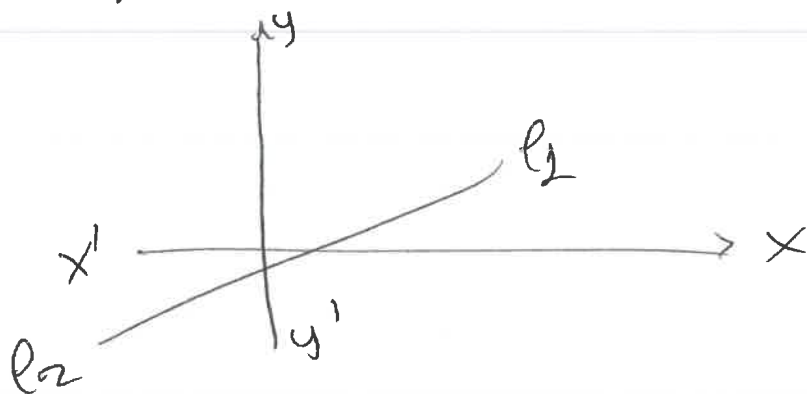
$$m_1 = m_2$$

( $l_1$  : line 1  
 $l_2$  : line 2)



Περίπτωση 3 | Οι εξισώσεις ταυτίζονται  
(Infinite Solutions)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad (\neq) \quad \begin{cases} m_1 = m_2 \\ b_1 = m_2 \end{cases}$$

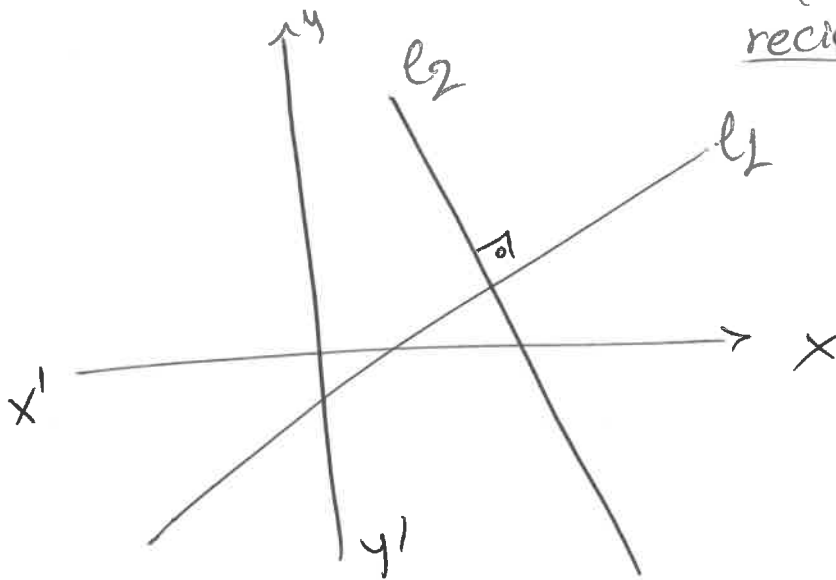


# (8) Perpendicular Lines (Καθήμες γραφές)

$$\begin{cases} (l_1) : y = m_1 \cdot x + b_1 \\ (l_2) : y = m_2 \cdot x + b_2 \end{cases}$$

$$\Rightarrow m_1 \cdot m_2 = -1$$

slopes are negative reciprocals of each other



## (9) Statistics

\* mean = average =  $\frac{\text{sum}}{\text{total}}$

\* median = middle number (when data is sorted from least to greatest)

\* mode = most frequent value

\* range = (greatest value) - (smallest value)

\* Standard Deviation (std) :

Measure of how spread out the data is.

- Low std  $\rightarrow$  values are close together
- High std  $\rightarrow$  values are spread apart

\* Margin of Error :

As sample increases ( $\uparrow$ )  $\Rightarrow$  margin of error decreases ( $\downarrow$ )

Known

Data Set A consists of the height of 60 buildings and has a mean of 30 meters. Data set B consists of the heights of 90 buildings and has a mean of 50 meters. Data set C consists of the heights of the 150 buildings from data sets A, B. What is the mean, in meters, of data set C?

Now

$$\text{dataset A: } \frac{\text{sum A}}{60} = 30 \Rightarrow \text{sum A} = 1800$$

$$\text{dataset B: } \frac{\text{sum B}}{90} = 50 \Rightarrow \text{sum B} = 4500$$

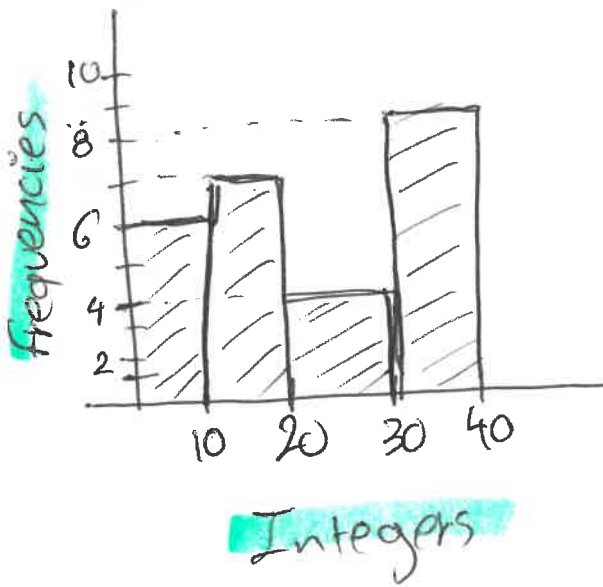
$$\text{Apa } \text{sum C} = \text{sum A} + \text{sum B} = 6300$$

$$\text{Apa } \text{mean C} = \frac{6300}{150} = 42$$

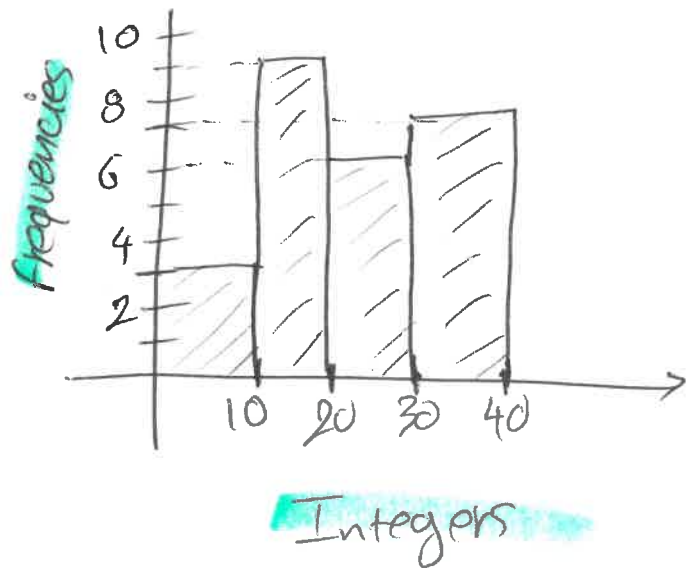
# Answer

Two data sets, each containing 25 integers, are displayed in the histograms below. For each histogram, the first interval represents the frequency of integers greater than or equal to 0 but less than 10, the second interval represents the frequency of integers greater than or equal to 10 but less than 20, and so on.

Data Set A



Data Set B



What is the smallest possible difference between the mean of data set A and the mean of data set B when they take their maximum/minimum values?

Λύση:

$$\text{Ουτίς} \rightarrow \text{mean} = \frac{\text{sum of (frequencies value)}}{\text{sum of (frequencies)}}$$

$$\max(\text{mean}(A)) = \frac{6 \cdot 0 + 7 \cdot 10 + 20 \cdot 4 + 8 \cdot 30}{6 + 7 + 4 + 8} = 15.6$$

$$\min(\text{mean}(A)) = \frac{6 \cdot 9 + 7 \cdot 19 + 20 \cdot 4 + 8 \cdot 39}{6 + 7 + 4 + 8} = 24.6$$

$$\min(\text{mean}(B)) = \frac{3 \cdot 0 + 9 \cdot 10 + 20 \cdot 6 + 7 \cdot 30}{3 + 9 + 6 + 7} = 16.8$$

$$\max(\text{mean}(B)) = \frac{3 \cdot 9 + 9 \cdot 19 + 20 \cdot 6 + 7 \cdot 39}{3 + 9 + 6 + 7} = 25.8$$

$$\text{Άρα: } \begin{cases} |\max(\text{mean}(A)) - \min(\text{mean}(B))| = 1.2 \\ |\max(\text{mean}(B)) - \min(\text{mean}(A))| = 1.2 \end{cases}$$

Όπως και να έχει, η απόσταση είναι 1.2.



# Ειδικοί Ψηφοί Ασκήσεων:

(1)

(1)  $15 \cdot x^{18} + x \cdot x^9 + 24$

The expression above has factor  $ax^9+b$  and  $c \cdot x^9+d$ ,  $a, b, c, d \in \mathbb{Z}$ . What is the maximum value of  $x$ ?

Λύση:

Από εκφώνηση:

$$15 \cdot x^{18} + x \cdot x^9 + 24 = (ax^9+b)(cx^9+d) \Rightarrow$$

$$\Rightarrow 15x^{18} + x \cdot x^9 + 24 = (a \cdot c)x^{18} + (a \cdot d + b \cdot c)x^9 + b \cdot d$$

$$\Rightarrow \begin{cases} a \cdot c = 15 \\ a \cdot d + b \cdot c = x \quad (1^*) \\ b \cdot d = 24 \end{cases}$$

Για να πραγματοποιηθεί η  $(1^*)$ , αρκεί να παρατηρήσω  
ότι πρέπει να πραγματοποιηθούν (έναν από τους δύο) οι  
όροι:  $a \cdot d$  ή  $b \cdot c$ .

Αυτό συμβαίνει όταν πάρω τις μικρότερες τιμές τα  $a, c$  και  $b, d$  βάσει των αρχικών συνθηκών.

Αν πάρω  $\begin{cases} a=15, c=1 \\ b=24, d=1 \end{cases}$  (i)  $\begin{cases} a=1, c=15 \\ b=1, d=24 \end{cases}$  τότε :

$$a \cdot d + b \cdot c = 1 + 15 \cdot 24 = \underline{\text{max value of } X}.$$

(2)  $f(x) = -(x+a)^2 \cdot (x-b) \cdot (x-c)$

$$\begin{cases} a, b, c > 0 \\ b < x < c \end{cases}$$

We have three statements :

(I)  $x > 0$

(II)  $f(x) > 0$

(III)  $f(0) > 0$

Which are true?

Λύση:

$$f(x) = 0 \Rightarrow \begin{cases} x = -a \text{ (δίνει } 0 \text{ για } a) \\ x = b \text{ (γιναι } 0 \text{ για } b) \\ x = c \text{ (-11-)} \end{cases} \Rightarrow$$

Ερωτήσεις:

$$f(0) = -a^2 \cdot (-b) \cdot (-c) = -a^2 \cdot bc < 0 \text{ (άρα το (III) } \underline{\text{όχι}} \text{ είναι αληθές)}$$

(3)

$$f(x) = -(x+a)^2 \cdot (x-b) \cdot (x-c)$$

και  $(x+a)^2 > 0 \Rightarrow -(x+a)^2 < 0$

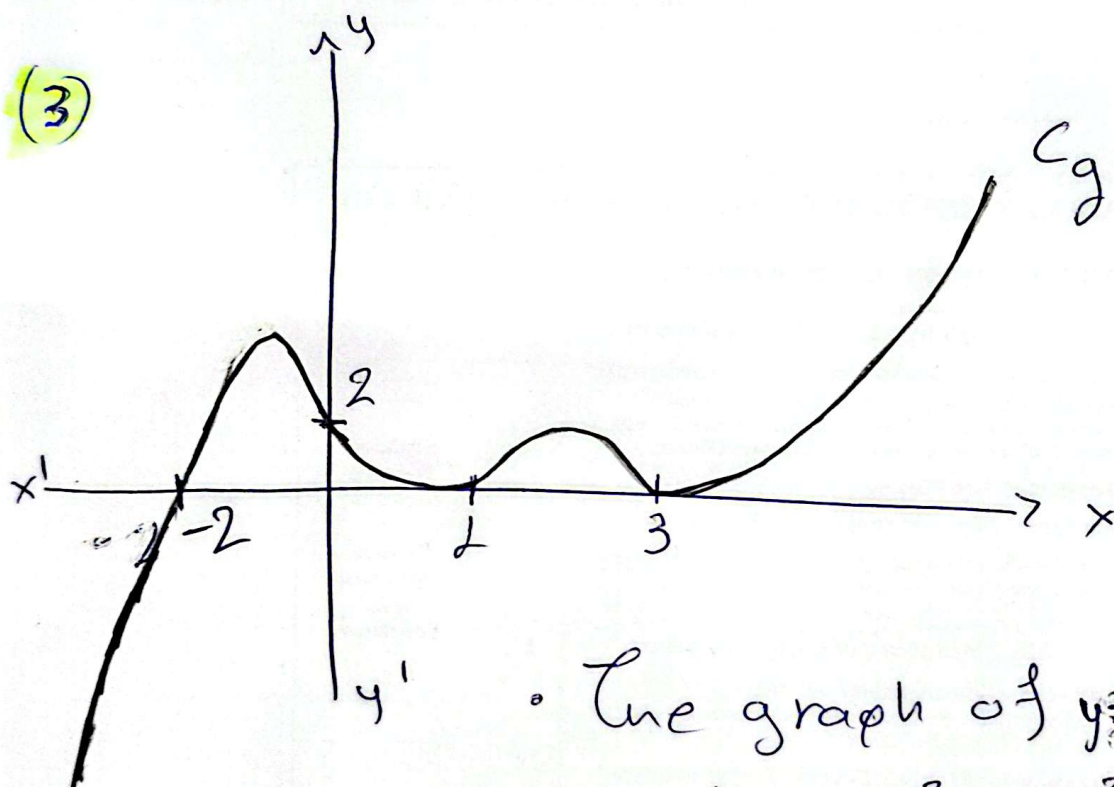
$$(x-b) > 0 \quad (x > b \text{ από υποθεση})$$

$$(x-c) < 0 \quad (x < c \text{ -- II --})$$

άρα  $f(x) > 0$  (το II)  $\therefore$  αληθεύει

Τέλος  $b, c > 0$  και  $x \in (b, c) \Rightarrow x > 0 \Rightarrow$  το (I) αληθεύει.

(3)



The graph of  $y = g(x)$  is shown.

If  $g(x+28) = \frac{1}{28} (x-a)^2 \cdot (x-b)^2 \cdot (x-c)$ , where

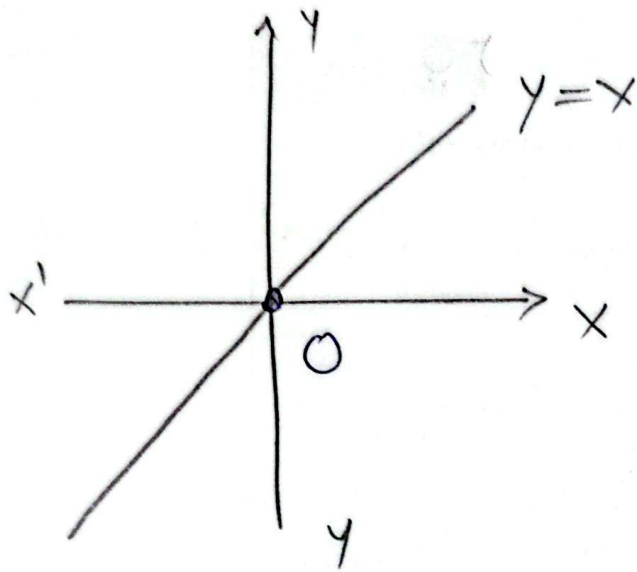
$a, b, c \in \mathbb{R}$ , find the ~~value~~ sign  $\overline{(a+b)^c}$ .  
(πρόσημο)



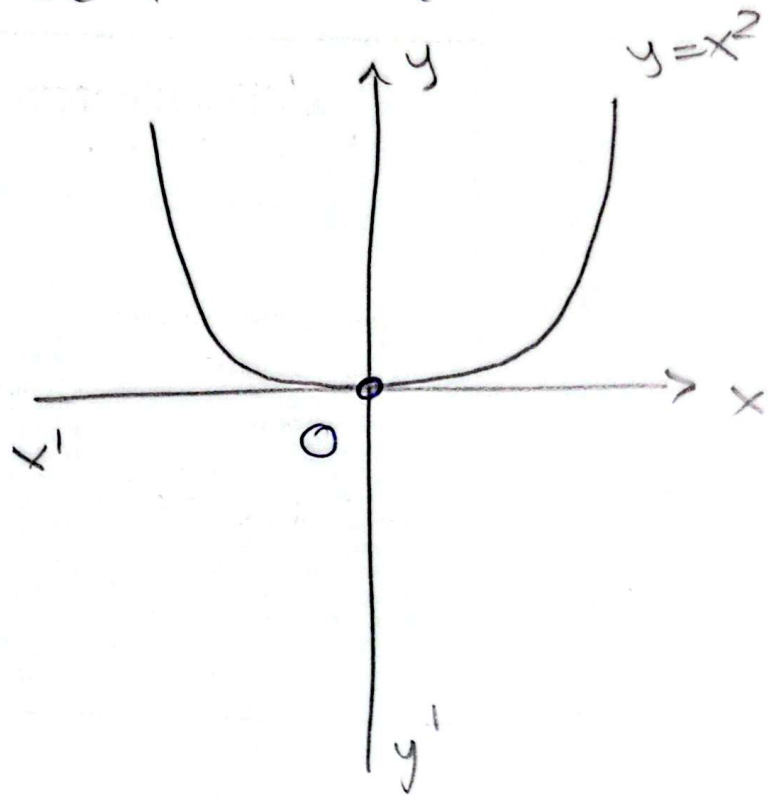
Λύση:

(4)

Αρχικά, θυμίσω τις εξής γραφικές παραστάσεις:



$$\boxed{f(x) = x}$$



$$\boxed{f(x) = x^2}$$

Παρατήρησε ότι γύρω από το  $x_0 = 0$ , η  $f(x) = x$  αλλάζει πρόσημο ( $f(x) < 0$  για  $x < 0$  και  $f(x) > 0$  για  $x > 0$ ) ενώ η  $f(x) = x^2$  διατηρεί πρόσημο ( $f(x) > 0$  για  $x \neq 0$ ).

Το τελεστικό είναι **χαρακτηριστική ιδιότητα** των πολυωνυμικών συναρτήσεων άρτια/περιττά βαθιά.

Συγκεκριμένα:

- Αν  $f(x) = x^{2k}$  (άρτια βαθιά) τότε διατηρεί πρόσημο γύρω από την ρίζα της
- Αν  $f(x) = x^{2k+1}$  (περιττά βαθιά) τότε αλλάζει πρόσημο γύρω από την ρίζα της.

Πίσω στην αρχική γραφική παράσταση, επειδή αλλάζει πρόσημο γύρω από το  $\underline{x_0 = -2}$  τότε αυτή αντιστοιχεί σε ρίζα του παράγοντα  $(x - x_0)^{(1)}$ , ενώ γύρω από τα  $\underline{x_0 = 1}$  και  $\underline{x_0 = 3}$  διατηρεί πρόσημο (κάνει καύτη), άρα αυτές αντιστοιχούν σε ρίζες παράγοντα της μορφής  $(x - x_0)^{(2)}$ . Άρα, από την υπόθεση, σκάγω το συντελεστή :

$$\begin{aligned} y = g(x) &= \frac{1}{28} (x-1)^2 \cdot (x-3)^2 \cdot (x-(-2)) = \\ &= \frac{1}{28} \underline{(x-1)^2} \cdot \underline{(x-3)^2} \cdot \underline{(x+2)} \end{aligned}$$



Το τελεστικό είναι **χαρκτηριστική ιδιότητα** των πολυωνυμικών συναρτήσεων άρτια/περιττά βαθιά.

### Συγκεκριμένα:

- Αν  $f(x) = x^{2k}$  (άρτια βαθιά) τότε διατηρεί πρόσημο γύρω από την ρίζα της
- Αν  $f(x) = x^{2k+1}$  (περιττά βαθιά) τότε αλλάζει πρόσημο γύρω από την ρίζα της.

Πίσω στην αρχική γραφική παράσταση, επειδή αλλάζει πρόσημο γύρω από το  $\underline{x_0 = -2}$  τότε αυτή αντιστοιχεί σε ρίζα του παράγοντα  $(x - x_0)^{(1)}$ , ενώ γύρω από τα  $\underline{x_0 = 1}$  και  $\underline{x_0 = 3}$  διατηρεί πρόσημο (κάνει κάρτηνα), άρα αυτές αντιστοιχούν σε ρίζες παράγοντα της μορφής  $(x - x_0)^{(2)}$ . Άρα, από την υπόθεση, σωστό το συμπέρασμα :

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Apa  $g = g(x+28) =$

(5)

$$= \frac{1}{28} ((x+28)-1)^2 \cdot ((x+28)-3)^2 \cdot ((x+28)+2) =$$

$$= \frac{1}{28} (x+27)^2 \cdot (x+25)^2 \cdot (x+30) =$$

$$= \frac{1}{28} (x - \underbrace{(-27)}_a)^2 \cdot (x - \underbrace{(-25)}_b)^2 \cdot (x - \underbrace{(-30)}_c)$$

Apa  $(a+b)^c = (-27-25)^{-30} = (-52)^{-30} =$

$$= \frac{1}{(-52)^{30}} = \frac{1}{52^{30} \cdot (-1)^{30}} = \frac{1}{52^{30}} > 0$$

(4)  $\begin{cases} 5x^2 + bx - 36 = (h \cdot x + x) \cdot (x + j) \\ h, x, j \in \mathbb{Z} \end{cases}$

Which of the following must be an integer?

(A)  $b/h$

(B)  $b/x$

(C)  $36/h$

(D)  $36/x$

Λύση:

(6)

Ανοίγω παρενθέσεις & κάνω πράξεις:

$$5x^2 + b \cdot x - 36 = h \cdot x^2 + (h \cdot j + k) \cdot x + k \cdot j \Rightarrow$$

$$\Rightarrow \begin{cases} h = 5 \\ b = h \cdot j + k \\ k \cdot j = 36 \Rightarrow \begin{cases} j = 36/k \in \mathbb{Z} \\ \text{or} \\ k = 36/j \in \mathbb{Z} \end{cases} \end{cases}$$

Άρα σωστό το (D).

(5) In the equation  $c \cdot x + j \cdot y = 10$ ,  $c, j \in \mathbb{R}$ . If the graph goes through the points  $(0, 5)$ ,  $(10, 7)$ , what's the value of  $c$  times  $j$ ?

Λύση:

- Για  $x=0, y=5$ :  $c \cdot 0 + j \cdot 5 = 10 \Rightarrow \underline{j=2}$
- Για  $x=10, y=7, j=2$ :  $c \cdot 10 + 2 \cdot 7 = 10 \Rightarrow \underline{c=-0.4}$
- Ζητάει το  $c \cdot j = 2 \cdot (-0.4) = \underline{-0.8}$



(6)

(7)

$$g(x) = c \cdot \sqrt{x} + d$$

$$c, d \in \mathbb{R}$$

$y = g(x)$  passes through the point  $(16, 0)$ .

$$g(36) < 0$$

Which of the following is true?

(A)  $g(0) = 36$

(B)  $g(0) = -36$

(C)  $c > d$

(D)  $c < d$

Lösung:

$$g(16) = 0 \Rightarrow c \cdot \sqrt{16} + d = 0 \Rightarrow 4 \cdot c + d = 0 \Rightarrow \underline{c = -\frac{d}{4}}$$

und

$$g(36) = c \cdot \sqrt{36} + d \stackrel{c = -\frac{d}{4}}{=} -\frac{d}{4} \cdot 6 + d = -\frac{3}{2} \cdot d + d =$$

$$= \underline{-\frac{1}{2}d} \text{ und } \underline{g(36) < 0} \Rightarrow$$

$$\Rightarrow -\frac{1}{2} \cdot d < 0 \Rightarrow \underline{d > 0} \text{ und an' en ox'ion}$$

$$c = -\frac{d}{4} \Rightarrow \underline{c < 0}. \text{ Aber } \underline{d > c}.$$

(7)  $f(x) = a \cdot \sqrt{x-b} + 12\sqrt{x-b}$

$$\begin{cases} f(4) = 0 \\ f(5) > 0 \end{cases}$$

(8)

Which could be a value for  $a+b$ ?

(A) -20

(B) -12

(C) -8

(D) -4

Given:

$$f(x) = \sqrt{x-b} \cdot (a+12) \left\{ \begin{array}{l} \underbrace{\sqrt{4-b}}_{\geq 0} \cdot (a+12) = 0 \end{array} \right. \quad (1*)$$

$$f(4) = 0$$

Έπειτα,  $\left| \underbrace{f(5) = \sqrt{5-b} \cdot (a+12)}_{\geq 0} > 0 \right| \quad (2*)$

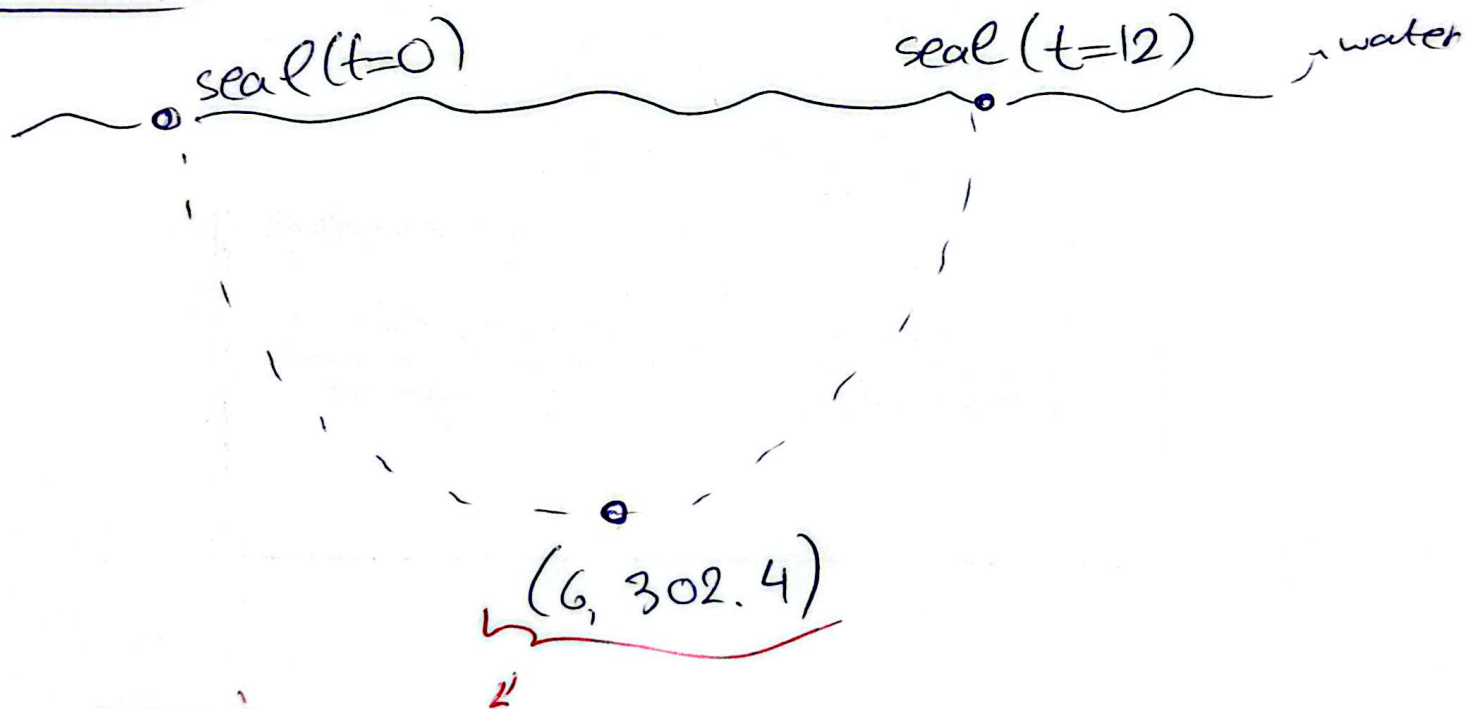
Answers (1\*), (2\*) :

$$\begin{cases} a+12 > 0 \\ \sqrt{4-b} = 0 \end{cases} \Rightarrow \begin{cases} a > -12 \\ |b| = 4 \end{cases} \Rightarrow$$

$$\Rightarrow a+b > -8 \Rightarrow \underline{(D) \text{ correct choice}}$$

(8) The quadratic function  $g$  models the depth, in meters, below the surface of the water of a seal  $t$  minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of water 12 minutes after it entered the water. Based on the function, what was the estimated depth to the nearest meter of the seal 10 minutes after it entered the water?

Solution:



korupiri  
(nāw te Vertex form)



$$y = a(x-h)^2 + k \Rightarrow$$

(10)

$$y = a(x-6)^2 + 302.4 \quad (1*)$$

και για  $x=12, y=0$  (ξαναβγαίνει στη επιφάνεια)

$$y \quad 0 = a(12-6)^2 + 302.4 \Rightarrow$$

$$a = - \frac{302.4}{36} = (-8.4) \quad (2*)$$

Από (1\*), (2\*)  $\Rightarrow$   $y = -8.4(x-6)^2 + 302.4$

Τέλος, βάσεις είναι  $x=10$  και υπολογίσεις...

(9)  $\begin{cases} f: \text{exponential} \\ f(1): \text{constant} \end{cases}$

(12)

Which of the following shows  $f(1)$  as the coefficient or the base?

- (A)  $f(x) = 50 \cdot (1.6)^{x+1}$
- (B)  $f(x) = 80 \cdot (1.6)^x$
- (C)  $f(x) = 128 \cdot (1.6)^{x-1}$
- (D)  $f(x) = 204.8 \cdot (1.6)^{x-2}$

Λύση:

Γενική μορφή exponential:  $y = M \cdot a^x$

$$y = M \cdot 1 = M$$

(σταθερά)

↙  $x=0$   
(ηδη έχει ο εξόρισ)

Άρα, βλέπεις σε ποιά αν'εις ελκδογίς το  $x=1$  η-  
δούλει ταν εκοίμ. (Σωστό είναι το (C)).

(3c)

•  $f, g$  : exponential

•  $\alpha, b \in \mathbb{Z} : \alpha < b$

•  $b < 0$

(I)  $f(x) = \alpha \cdot (2.1)^{x+b}$

(II)  $g(x) = \alpha \cdot (2.1)^x + b$

Which of the two displays the y-coordinate of the y-int of the corresponding function?

Solution:

~~Answer~~

\*  $\left\{ \begin{array}{l} f(0) = \alpha \cdot (2.1)^b \text{ και } (2.1)^b \neq 1 \text{ (άρα δεν είναι ο } y\text{-int)} \\ g(0) = \alpha \cdot 1 + b, \text{ άρα δεν είναι } \alpha \cdot 1 \text{ (άρα δεν είναι ο } y\text{-int)} \end{array} \right.$

Καίρια αν'τα δύο.



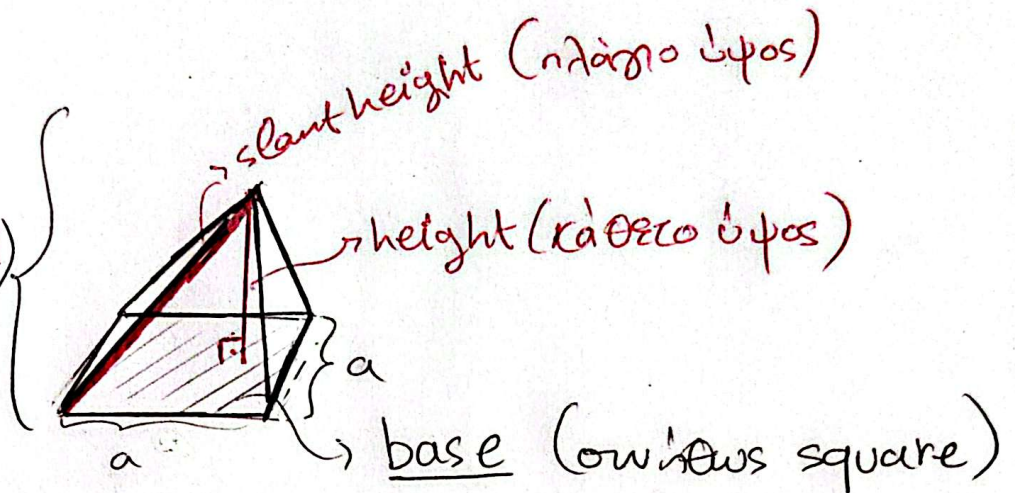
## (1) Pyramids

Γενικό Σχήμα

$$V = \frac{1}{3} B \cdot h \quad (\text{volume})$$

(B: επιφάνεια βάσης)

(h: κάθετος ύψος)

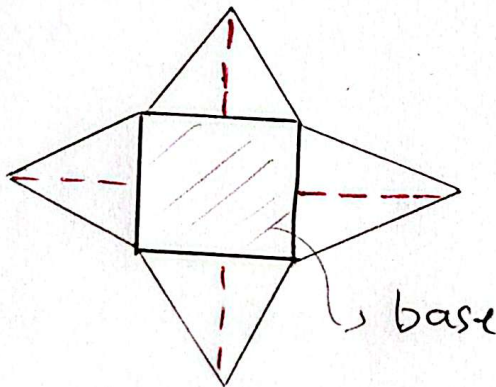


Surface Area

$$S = A + \frac{1}{2} p \cdot s$$

όπου  $\begin{cases} A = \text{Area of base} \\ p = \text{perimeter of base} \\ s = \text{slant height} \end{cases}$

Intuition  
για το  
τίπο



--- : slant height

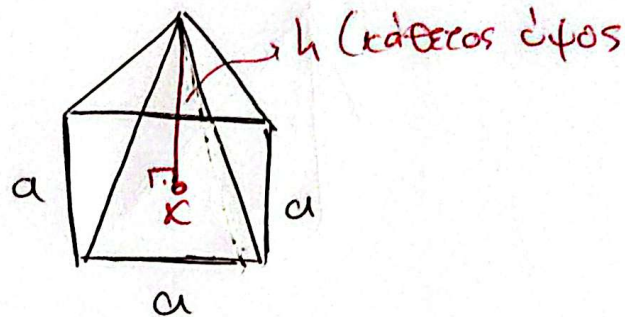
Επίλυση Άσκησης:

Consider a square right pyramid with volume = 400 cube inch (inch<sup>3</sup>) and a slant height of 13 inches. The pyramid's length, height are integers. What is the surface area of the pyramid in (inch)<sup>2</sup>?

Λύση:

$$(\sum x_i^2 - a)$$

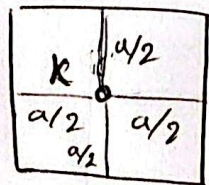
(2)



Έστω  $a$ : πλευρά της τετραγωνικής βάσης.

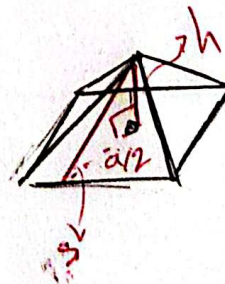
Το σημείο  $K$  (εκεί που προβάλλεται η κορυφή της πυραμίδας) βρίσκεται στο κέντρο της τετραγωνικής βάσης.

Αρα:



Για να βρω το κάθετο ύψος  $h$  κάνω πυθαγόρειο στο τρίγωνο με πλευρές  $a/2, h, s$ :

κάτοψη τριγώνου →





(3)

$$(a/2)^2 + h^2 = s^2 \Rightarrow$$

$$(a/2)^2 + h^2 = 13^2 \quad (1*)$$

Σερα :

$$V = \frac{1}{3} B \cdot h$$

$$B = a^2$$

$$V = 400 \text{ (inch)}^3$$

$$\frac{1}{3} a^2 \cdot h = 400 \Rightarrow$$

$$\Rightarrow |a^2 \cdot h = 1200| \quad (2*)$$

$$(\Leftrightarrow a^2 = 1200/h)$$

Ανò (1\*), (2\*) :

~~$$\frac{a^2}{4} + \frac{1200}{a^2} = 13^2$$~~

~~$$a = 20$$~~

~~$$h^2 = 69 - 100$$~~

~~$$h^2 = 69$$~~

~~$$h = \sqrt{69}$$~~

~~$$\frac{1200}{h} + h^2 = 13^2 \Rightarrow$$~~

~~$$\frac{1200}{4h} + h^2 = 13^2 \Rightarrow 4 \cdot h^3 = (13^2 \cdot 4) \cdot h + 1200 \quad (3)$$~~

(4)

$$\begin{cases} y = \frac{1200}{x^2} \\ \frac{x^2}{4} + y^2 = 169 \end{cases} \Rightarrow$$

$$\left( \begin{array}{l} \text{titik potong: } u=y \\ x=a \end{array} \right)$$

$$\Rightarrow \frac{x^2}{4} + \frac{(1200)^2}{x^4} = 169 \quad \cdot x^4 \Rightarrow$$

$$\Rightarrow \frac{x^6}{4} - 169 \cdot x^4 + (1200)^2 = 0 \Rightarrow$$

$$\Rightarrow x^4 \left( \frac{x^2}{4} - 169 \right) = -1200^2 \Rightarrow$$

$$\Rightarrow x^4 \left( \left( \frac{x}{2} \right)^2 - 13^2 \right) = -(3 \cdot 4 \cdot 100)^2$$

$$\Rightarrow \underbrace{x^4}_{10^4} \cdot \underbrace{\left( \frac{x}{2} - 13 \right)}_{(-1)} \cdot \underbrace{\left( \frac{x}{2} + 13 \right)}_{3} = \underbrace{10^4}_{10^4} \cdot \underbrace{(-1)}_{(-1)} \cdot \underbrace{3}_{3} \cdot \underbrace{4}_{4}$$

$$\Rightarrow x=10 \Rightarrow \boxed{a=10}$$

Area:  $S = A + \frac{1}{2} p \cdot s =$   
 $= 10^2 + \frac{1}{2} \cdot (4 \cdot 10) \cdot 13$   
 $= 100 + 260 = \underline{360}$



## (2) Probabilities (Independent Events)

### Review Apx 1:

If events A, B are independent (the result of one of them, does not affect the outcome of the other) then :

$$P(\underbrace{A \text{ and } B}) = \underbrace{P(A) \cdot P(B)}$$

probability the  
A and B happen  
simultaneously

probability of event A  
times the probability of  
event B

### Exmp 1:

A high school math club consists of 12 members :

- 5 freshmen
- 4 sophomores
- 3 juniors

A committee of 3 members is to be randomly selected to organize an upcoming math camp. What is the probability that the committee will include ~~at least~~ one member from each grade?



Λύση

(6)

$$\text{total} = 5 + 4 + 3 = 12$$

Επιτροπή

1<sup>η</sup> Ομάδα  
(juniors)

2<sup>η</sup> Ομάδα  
(sophomore)

3<sup>η</sup> Ομάδα  
(freshmen)

$$p(1 \text{ jun}, 1 \text{ soph}, 1 \text{ fresh}) =$$

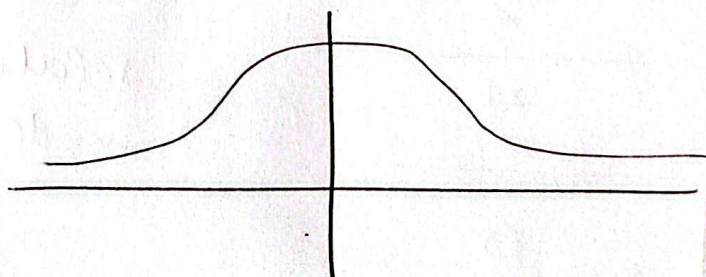
$$= p(1 \text{ jun}) \cdot p(1 \text{ soph}) \cdot p(1 \text{ fresh}) =$$

$$= \frac{3}{12} \cdot \frac{4}{12} \cdot \frac{5}{12} = \frac{5}{12^2} = \frac{5}{144}$$

(3) Symmetric Sets :

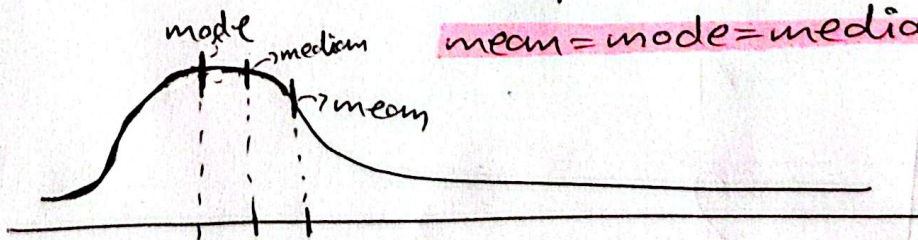
3 είδη κατανομών :

(A) Συμμετρική



$$\text{mean} = \text{mode} = \text{median}$$

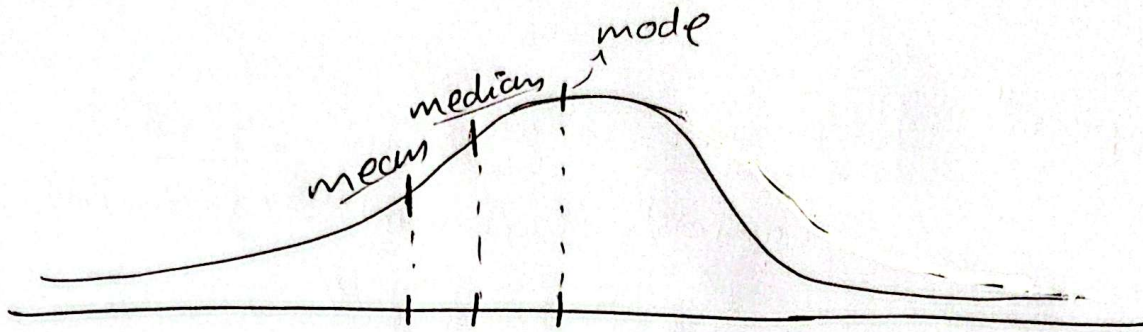
(B) Positive  
Skew



$$\text{mode} < \text{median} < \text{mean}$$



## (7) Negative Skew



$$\text{mean} < \text{median} < \text{mode}$$

### Expmion

Let the following be a distribution frequency table for data set A:

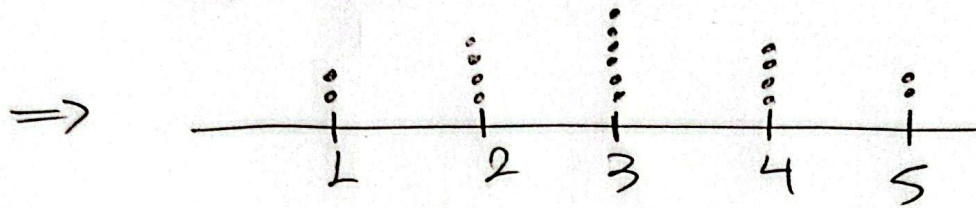
| <u>value</u> | <u>frequency</u> |
|--------------|------------------|
| 1            | 2                |
| 2            | 4                |
| 3            | 6                |
| 4            | 4                |
| 5            | 2                |

• What is the relationship between mean, mode, median?

Qion

(8)

Set A is symmetric  $\Rightarrow$



$\Rightarrow$  mean = mode = median.

#### (4) Unit Conversions

Exercises

If a car accelerates at  $108 \text{ km/hour}^2$ , find the acceleration in  $\text{m/s}^2$ :

$$108 \frac{\text{km}}{(\text{hour})^2} = 108 \cdot \frac{1000 \text{ m}}{(3600 \text{ sec})^2} =$$

$$= 108 \cdot \frac{1000}{6^2 \cdot 10^4} \frac{\text{m}}{\text{s}^2} =$$

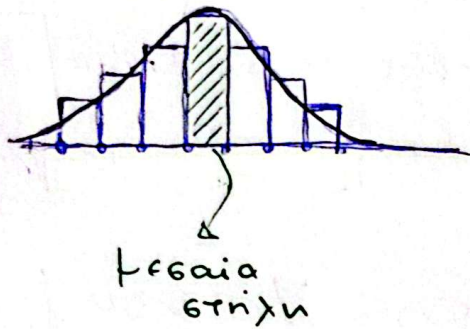
$$= 18 \cdot \frac{1}{60} \frac{\text{m}}{\text{s}^2} = \frac{9}{30} \frac{\text{m}}{\text{s}^2}$$



# Είδος Συμμετρίας !

(9)

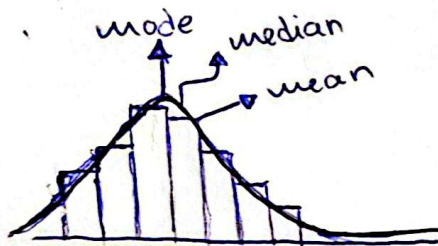
①



Symmetric Dataset:

$$\left. \begin{array}{l} \text{mean} \\ = \text{mode} \\ = \text{median} \end{array} \right\} \text{ταυτιζόμενα}$$

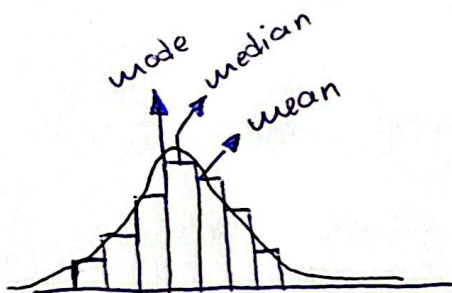
②



Positive skewness:

$$\text{mode} < \text{median} < \text{mean}$$

③



Negative skewness:

$$\text{mean} < \text{median} < \text{mode}$$



A cyclist travels from town A to town B, a distance of 120 kilometers, at a constant speed. If the cyclist increases the speed by 10 km/h, the journey would take one hour less. What was the original speed of the cyclist in kilometers per hour?

- A) 30
- B) 35
- C) 40
- D) 50

SOS Αποκλιση  
Φυσική

Solution:

$$\left. \begin{aligned} 120 &= v \cdot t \\ 120 &= (v+10)(t-1) \end{aligned} \right\} \rightarrow v \cdot t - v + 10t - 10 = v \cdot t$$

$$v = 10t - 10$$

$$\Rightarrow (10t - 10) \cdot t = 120$$

$$10t^2 - 10t - 120 = 0$$

$$t^2 - t - 12 = 0$$

$$\Rightarrow t = 4, t = 3$$

$$\rightarrow v = 30 \text{ km/h.}$$

(Ευχαριστώ Νικόλα Μεταξά)

## (5) Exponentials

(11) (9)

(A)  $g(x) = (0.48)^{x/4}$  is re-written as  
 $g(x) = \left(1 - \frac{n}{100}\right)^x$

find n :

$$g(x) = ((0.48)^{1/4})^x = (0.83)^x = \left(1 - \frac{17}{100}\right)^x \Rightarrow$$
$$\Rightarrow \underline{n=17}$$

(B) How many years will it take for a population of bacteria growing at an annual rate of 50% to quadruple in size?

Given:  $y = f(x) = P_0 \cdot \left(\frac{3}{2}\right)^x$  ( $P_0$  = initial population)

$$\Rightarrow 4 \cdot P_0 = P_0 \left(\frac{3}{2}\right)^x \Rightarrow \left(\frac{3}{2}\right)^x = 4 \Rightarrow$$

$$\left(\frac{3}{2}\right)^2 = 2.25$$

$$\Rightarrow 3 < x < 4 \Rightarrow \text{approximately 3.5 years}$$

$$\left(\frac{3}{2}\right)^3 = 3.375$$

$$\left(\frac{3}{2}\right)^4 = 5.0625$$



$$(r) \begin{cases} f(x) = a \cdot b^{x/4} \\ f(2) = 6 \\ f(5) = 162 \end{cases}$$

find  $f(7)$

(12)

Solution:

$$\left. \begin{aligned} f(2) = 6 &\Rightarrow a \cdot (b^{1/4})^2 = 6 \quad (1*) \\ f(5) = 162 &\Rightarrow a \cdot (b^{1/4})^5 = 162 \quad (2*) \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow \frac{(2*)}{(1*)} = \frac{a \cdot (b^{1/4})^5}{a \cdot (b^{1/4})^2} = \frac{162}{6} \Rightarrow$$

$$\Rightarrow (b^{1/4})^3 = 27 \Rightarrow (b^{1/4})^3 = 3^3 \Rightarrow$$

$$\Rightarrow \underline{b^{1/4} = 3} \Rightarrow \underline{a = \frac{2}{3}} \Rightarrow$$

$$\Rightarrow f(x) = \frac{2}{3} \cdot 3^x = 2 \cdot 3^{x-1}$$

then substitute to  $f(7) = 2 \cdot 3^{7-1} = \dots$



## (6) Radical Expressions

(13)

$$f(x) = \frac{x^2 + a \cdot x + b}{2 \cdot x + c}$$

•  $f(x)$  does not intersect line  $x=4$

•  $f(6) = 0$

•  $f(8) = 0$

find  $a+b+c$  :

• 
$$\left. \begin{array}{l} 2x + c = 0 \\ x = 4 \end{array} \right\} \Rightarrow 8 + c = 0 \Rightarrow \underline{c = -8}$$

• 
$$\left. \begin{array}{l} f(6) = 0 \\ f(8) = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} 36 + 6a + b = 0 \\ 64 + 8a + b = 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} a = -14 \\ b = 48 \end{array} \right.$$

•  $a + b + c = -14 + 48 - 8 = 26$



(7) Completing the square  
(Circle Equation)

14

$$\begin{cases} x^2 - 4x + y^2 - 8y - 80 = 0 \\ \text{find the circumference} \end{cases}$$

Solution

$$x^2 - \underline{2} \cdot \cancel{x} \cdot \underline{2} + \underline{2}^2 + y^2 - \underline{2} \cdot 4y + \underline{4}^2 = 80 + \underline{2}^2 + \underline{4}^2$$

$$(x-2)^2 + (y-4)^2 = 10^2 \Rightarrow$$

$$\Rightarrow \begin{cases} C(2, 4) : \text{center} \\ r = 10 \end{cases}$$

$$\Rightarrow \text{circ} = 2\pi r = \underline{20\pi}$$



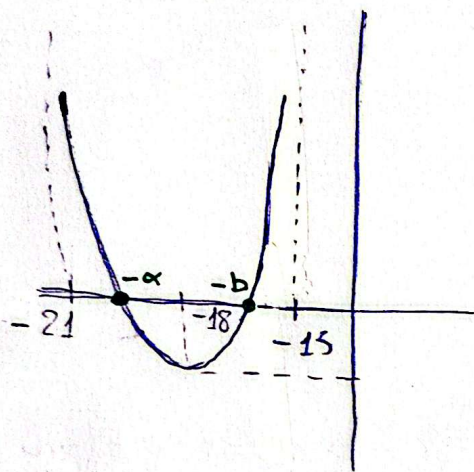
## SOS - Άγνωστο!

(15)

$$\begin{cases} f(x) = (x+a)(x+b) \\ a, b \in \mathbb{Z} \\ f(-21) > 0 \\ f(-15) > 0 \\ f(-18) < 0 \end{cases}$$

Possible value for  $a+b$ ?

Solution:



$$\begin{cases} -21 < a < -18 \\ -18 < -b < -15 \end{cases} \Rightarrow \begin{cases} 18 < a < 21 \\ 15 < b < 18 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} a \in \{19, 20\} \\ b \in \{16, 17\} \end{cases}$$

πιθανά

αθροίσματα:  $\cdot 19 + 16$

$\cdot 19 + 17$

$\cdot 20 + 16$

$\cdot 20 + 17$

## SOS - άσκηση #2 !

16

3 points in xy-plane

A (1,0)

B (0,0)

C (-1,0)

What is the measure of angle  $\angle abc$ ?

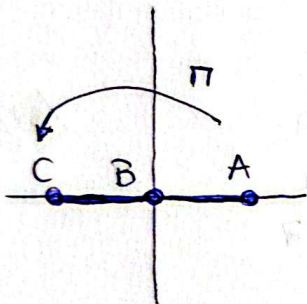
A)  $456 \cdot \pi/6$

B)  $459 \cdot \pi/6$

C)  $462 \cdot \pi/6$

D)  $468 \cdot \pi/6$

Solution:



$$(A): \frac{456\pi}{6} = 76 \cdot \pi = 2\pi \cdot 38 \rightarrow \text{ανόρισππτετα}$$

$$(B): \frac{459\pi}{6} = 76 \cdot \pi + \frac{\pi}{2} \rightarrow \text{ανόρισππτετα}$$

$$(C): \frac{462\pi}{6} = 77\pi = 76\pi + \pi = 2\pi \cdot 38 + \pi = \pi$$

δεκτπ

credits to math!